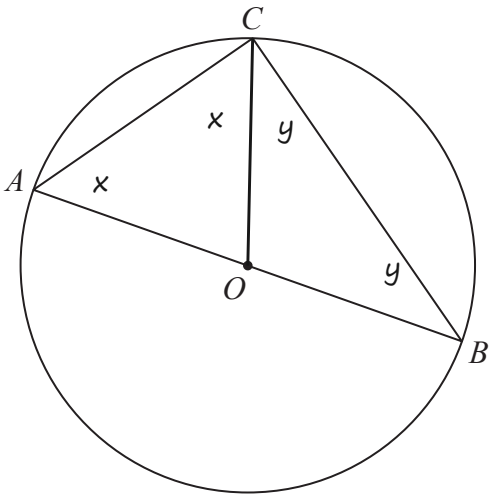
20



A, B and C are points on the circumference of a circle, centre O.
AOB is a diameter of the circle.

Prove that angle ACB is 90°
You must **not** use any circle theorems in your proof.

Draw OC. OA = OB = OC (radii).

1 mark
Construct OC and use equal radii

Triangles AOC and BOC are isosceles.

Their base angles are equal:

OAC = ACO = x; OBC = BCO = y.

In triangle ABC: x + (x + y) + y = 180°.

1 mark
Angles in a triangle sum to 180°

2x + 2y = 180°, so x + y = 90°.

1 mark
Complete the angle calculation

Angle ACB = x + y = 90°.

1 mark
Complete proof: equal radii, equal base angles and triangle angle sum

(Total for Question 20 is 4 marks)